# Convolutional Neural Network Classification of Simulated LArTPC Images for Supernova Neutrino Detection

## 1 Introduction

When a massive star exhausts its nuclear fuel, the resulting gravitational collapse releases approximately $3 \times 10^{53}$ erg of energy, of which around 99% is emitted as a burst of neutrinos with energies of order 10 MeV over a timescale of tens of seconds [1, 2]. The only supernova from which neutrinos have been directly detected is SN1987A, observed on 23 February 1987, when multiple detectors including Kamiokande-II recorded a total of 20 neutrino events from the Large Magellanic Cloud, providing the first confirmation of core-collapse neutrino emission [2].

Liquid argon time projection chambers (LArTPCs) represent a next-generation detection technology, reconstructing three-dimensional charged-particle trajectories from high-resolution two-dimensional projections collected on wire readout planes [3]. However, the resulting images are highly sparse and subject to multiple background sources, including electronic readout noise and radioactive contamination from isotopes such as $^{39}$Ar naturally present in liquid argon [3, 4], making reliable event identification a significant challenge.

Convolutional neural networks (CNNs) are well suited to this task, as they are capable of learning spatially structured features directly from image data without manual feature engineering [3]. This work investigates the feasibility of CNN-based binary classification of simulated LArTPC images, distinguishing neutrino signal events from pure noise backgrounds under both electronic and radioactive noise conditions.

## 2 Method

### 2.1 Dataset

The dataset consists of 10,000 simulated LArTPC images, each of 100×100 pixels, where pixel values represent energy deposited in the detector within a given spatial and temporal slice. The statistical properties of the raw neutrino images were examined prior to noise simulation to inform the choice of noise parameters. Analysis reveals extreme sparsity, with only 0.47% of pixels carrying non-zero values. Among non-zero pixels, the mean value is 4.66 with a standard deviation ($\sigma$) of 5.83. This scale directly motivated the choice of noise levels explored in Section 2.2, where a noise standard deviation of the same order of magnitude represents a physically relevant benchmark. Representative neutrino images and the pixel value distributions are shown in Figure 1.

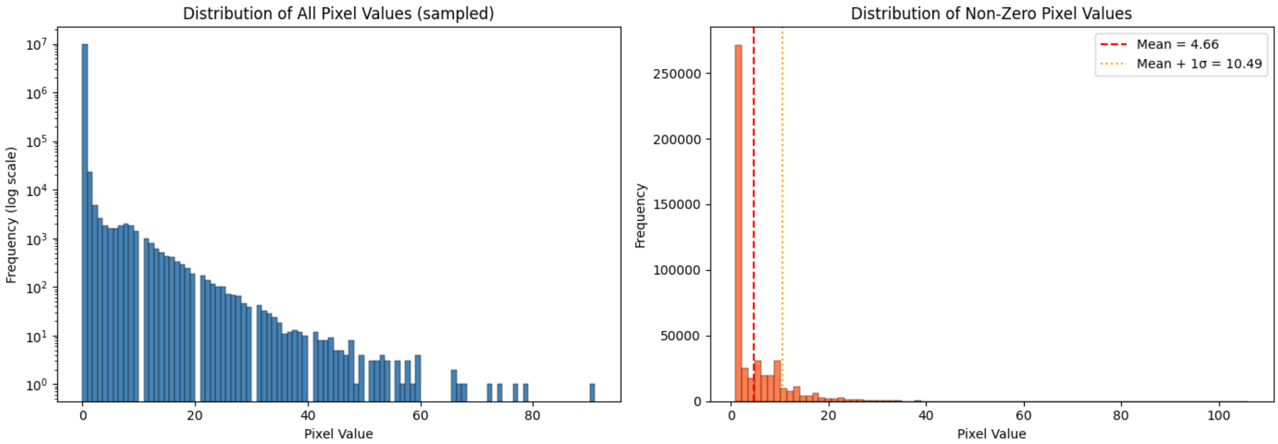



Figure 1: Pixel value distributions of the raw neutrino image dataset. Left: full distribution on a logarithmic scale, showing extreme sparsity. Right: distribution of non-zero pixels only, with mean and mean+$1\sigma$ indicated by dashed lines.

| Metric | Value |
|---|---|
| Total pixels | 100,000,000 |
| Non-zero pixels | 473,519 (0.47%) |
| Non-zero mean | 4.66 |
| Non-zero std | 5.83 |
| Non-zero max | 106.00 |

Table 1: Summary of Pixel Statistics

## 2.2 Electronic Noise Simulation

Electronic noise was simulated by drawing pixel values independently from a zero-mean Gaussian distribution $N(0, \sigma^2)$, consistent with the thermal noise model used in the detector simulation. This is physically motivated, as electronic readout noise in LArTPC detectors is well described by a Gaussian process. A range of noise levels was investigated, with $\sigma$ varying from 1 to 15 , covering conditions well below to well above the signal pixel scale identified in Section 2.1.

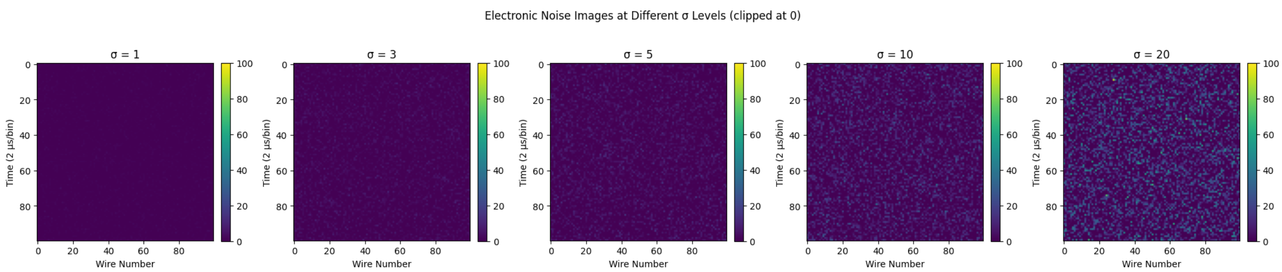



Figure 2: Representative noise images at $\sigma \in \{1, 3, 5, 10, 20\}$, clipped at zero.

Noise images were generated as raw Gaussian arrays retaining both positive and negative values. When constructing signal images, the noise was first overlaid onto the neutrino image and the result was then clipped at zero. Pure noise background images were generated and clipped independently. This ordering ensures that negative noise fluctuations can partially cancel signal pixels, as would occur physically. Representative noise images at several $\sigma$ values are shown in Figure 2, and a direct comparison between a clean neutrino image, a pure noise image, and a neutrino-plus-noise image at

$\sigma = 5$ is shown in Figure 3.

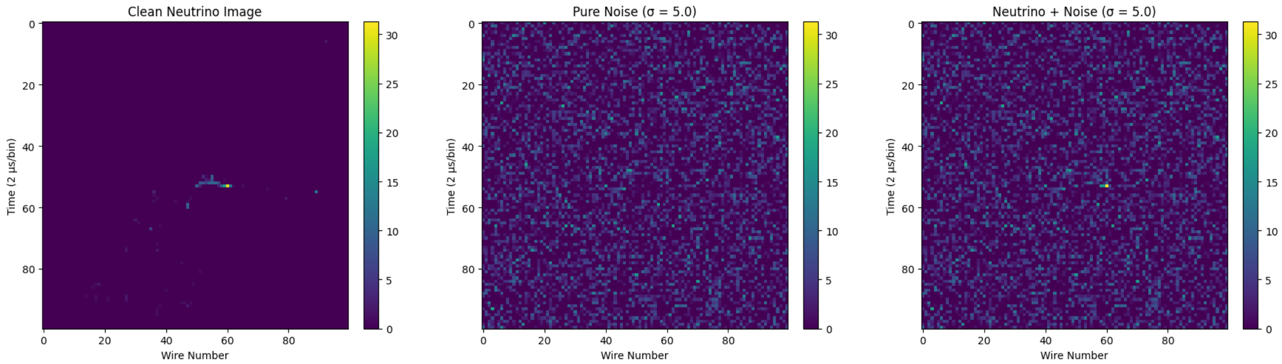



Figure 3: Comparison of a clean neutrino image (left), a pure electronic noise image at $\sigma = 5$ (centre), and the corresponding neutrino-plus-noise image (right).

### 2.3 CNN Classifier Design and Training

A balanced binary classification dataset was constructed from the neutrino image pool and synthetically generated noise images. Signal samples, assigned label 1, were formed by overlaying electronic noise onto neutrino images and clipping at zero. Background samples, assigned label 0, consisted of pure noise images clipped at zero. The combined dataset was normalised globally using min-max scaling, $X' = \frac{X - X_{min}}{X_{max} - X_{min}}$, where $X$ is the original value, $X'$ is the scaled value, $X_{min}$ and $X_{max}$ are the minimum and maximum values of the feature respectively. A channel dimension was appended to each image to match the CNN input format of $(100, 100, 1)$. Representative examples of normalised signal and background images are shown in Figure 4.

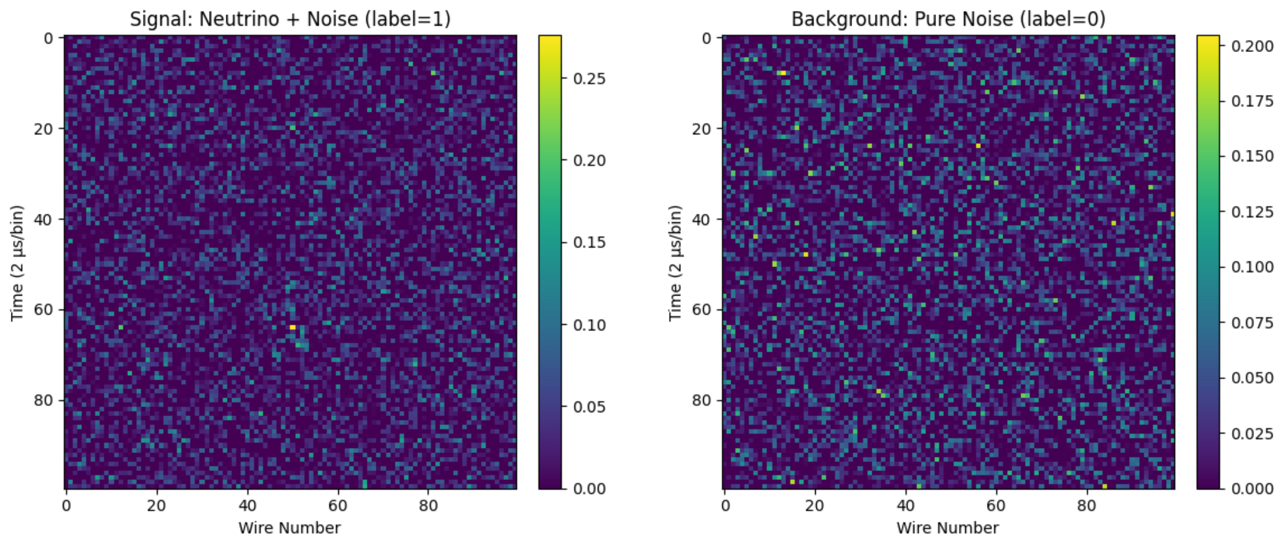



Figure 4: Representative normalised images from the classification dataset at $\sigma = 5$. Left: signal sample (neutrino overlaid with electronic noise, label 1). Right: background sample (pure noise, label 0). Pixel values are scaled to $[0, 1]$ via global min-max normalisation.

The 10,000 neutrino images were randomly permuted and partitioned into three non-overlapping pools: 5,000 for training, 1,500 for validation, and 1,500 for testing. Background noise images were generated independently for each split, ensuring no image contributed to more than one set.

#### 2.3.1 Network Architecture

The classifier was implemented as a CNN designed to process $100 \times 100$ single-channel images. The architecture consists of three convolutional blocks, each comprising a Conv2D layer with a $3 \times 3$ kernel, ReLU activation, and no zero-padding, followed by a MaxPooling2D layer with pool size $2 \times 2$. The number of filters was set to 16 in the first block and 32 in the second and third blocks, keeping the initial capacity low to avoid overfitting to the background Gaussian noise patterns. The convolutional output was flattened and passed through a Dense layer of 64 units with ReLU activation, followed by a Dropout layer with rate 0.6 to regularise the network and encourage the learning of robust features. The output layer consisted of a single unit with sigmoid activation, appropriate for binary classification. The architecture is summarised in Table 2.

| **Layer** | **Output Shape** | **Parameters** |
|---|---|---|
| Conv2D (16 filters, $3 \times 3$, ReLU) | $98 \times 98 \times 16$ | 160 |
| MaxPooling2D ($2 \times 2$) | $49 \times 49 \times 16$ | 0 |
| Conv2D (32 filters, $3 \times 3$, ReLU) | $47 \times 47 \times 32$ | 4,640 |
| MaxPooling2D ($2 \times 2$) | $23 \times 23 \times 32$ | 0 |
| Conv2D (32 filters, $3 \times 3$, ReLU) | $21 \times 21 \times 32$ | 9,248 |
| MaxPooling2D ($2 \times 2$) | $10 \times 10 \times 32$ | 0 |
| Flatten | 3,200 | 0 |
| Dense (64, ReLU) | 64 | 204,864 |
| Dropout (rate = 0.6) | 64 | 0 |
| Dense (1, Sigmoid) | 1 | 65 |
| **Total trainable parameters** | | 218,977 |

Table 2: CNN architecture used for binary classification of LArTPC images.

#### 2.3.2 Training

The model was trained using the Adam optimiser with a learning rate of $1 \times 10^{-4}$ and binary cross-entropy loss. Two callbacks were employed during training. EarlyStopping monitored validation loss with a patience of 6 epochs and restored the best weights upon termination, while ReduceLROnPlateau reduced the learning rate by a factor of 5 when validation loss showed no improvement for two consecutive epochs. Before each training run, model weights were reloaded from a saved initial checkpoint and the model was recompiled to ensure consistent initialisation across runs.

Classifier performance was evaluated using four metrics. Letting TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives respectively, the metrics are defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Efficiency} = \frac{TP}{TP + FN} \tag{2}$$

$$\text{Purity} = \frac{TP}{TP + FP} \tag{3}$$

where efficiency measures the fraction of genuine neutrino events correctly identified, and purity measures the fraction of positive predictions that correspond to genuine neutrino events. The area under the ROC curve (AUC) provides a threshold-independent measure of discriminative power. A systematic scan over classification thresholds from 0.05 to 0.95 was conducted to characterise the efficiency–purity trade-off, with the default threshold set to 0.5.

The model was trained independently at each of eight values of $\sigma_{\text{train}} \in \{1, 3, 5, 6, 7, 9, 10, 15\}$, with weights and optimiser state fully reset before each run. Each trained model was then evaluated on test sets generated at all eight values of $\sigma_{\text{test}} \in \{1, 3, 5, 6, 7, 9, 10, 15\}$, producing a complete $8 \times 8$ performance matrix. This design allows the generalisation behaviour of each model to be assessed under both matched ($\sigma_{\text{test}} = \sigma_{\text{train}}$) and mismatched conditions.

## 2.4 Radioactive Noise Simulation

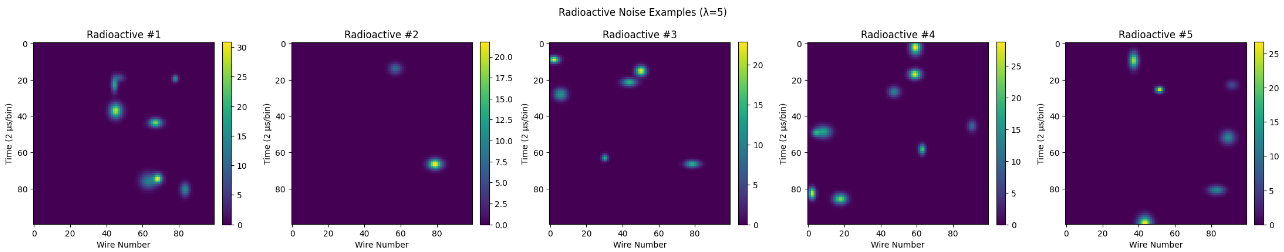



Figure 5: Representative radioactive noise images at $\lambda = 5$, showing 5 independently generated samples. Each image contains a Poisson-distributed number of 2D Gaussian blobs with randomised position, amplitude, and spatial width.

Liquid argon detectors are susceptible to contamination by naturally occurring radioisotopes, most notably $^{39}$Ar. Unlike electronic noise, which introduces spatially uniform pixel-level fluctuations, radioactive noise manifests as spatially confined signals. This structural similarity to genuine neutrino interactions makes radioactive noise a different background from the electronic noise.

Radioactive noise images were simulated by placing a random number of two-dimensional Gaussian blobs onto an otherwise empty $100 \times 100$ pixel canvas. The number of blobs per image was drawn from a Poisson distribution with mean $\lambda$, which serves as the controlling parameter analogous to $\sigma$ in the electronic noise case. Each blob was modelled as a 2D Gaussian of the form

$$f(x, y) = A \exp\left(-\frac{(x - x_0)^2}{2\sigma_x^2} - \frac{(y - y_0)^2}{2\sigma_y^2}\right), \tag{4}$$

where $(x_0, y_0)$ is the centre position sampled uniformly across the image, $A$ is the peak amplitude drawn uniformly from $[5, 30]$, and $\sigma_x$, $\sigma_y$ are the spatial widths along the wire and time axes respectively, each drawn independently and uniformly from $[1, 3]$ pixels.

The same CNN architecture and training procedure described in Section 2.3 were applied without modification. A dedicated model was trained independently at each of seven values of $\lambda \in \{1, 5, 10, 50, 100, 500, 1000\}$ with weights and optimiser state fully reset between runs. Performance was evaluated on a test set generated at the corresponding $\lambda$ value, and the results were compared directly against those obtained for electronic noise to assess which noise type poses a greater challenge to the classifier.

# 3 Results

## 3.1 Baseline Classifier

The CNN was first trained and evaluated at the physical baseline noise level of $\sigma = 5$, corresponding to the electronic noise standard deviation used in the detector simulation. The training and validation curves are shown in Figure 6, where both accuracy and loss converge smoothly within 20 epochs. The loss curves revealed a modest gap between training and validation loss in later epochs, indicative of mild overfitting; however, validation accuracy remained stable throughout, suggesting that the model retained adequate generalisation ability.

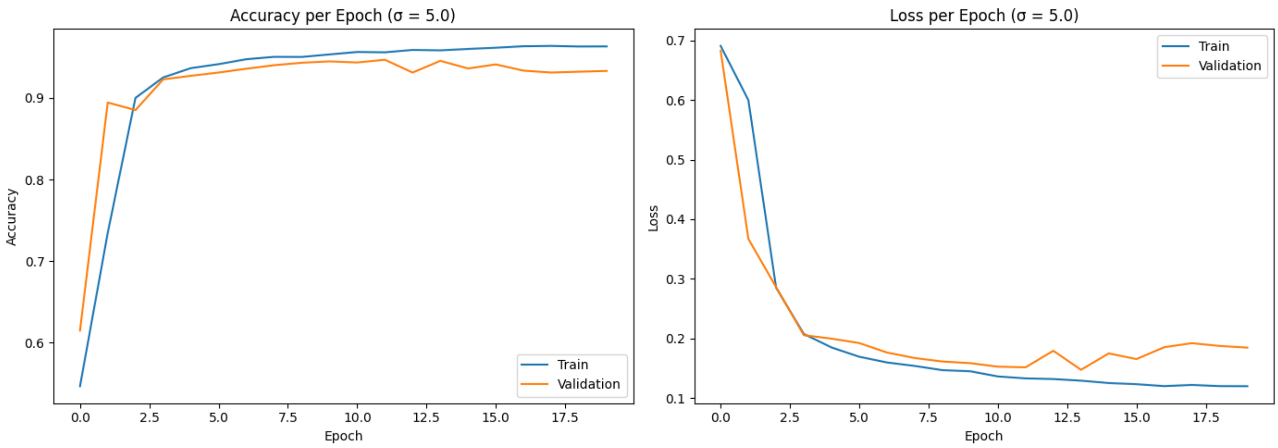



Figure 6: Training and validation curves for the baseline classifier at $\sigma = 5$. Left: accuracy per epoch. Right: binary cross-entropy loss per epoch. Early stopping was triggered after approximately 20 epochs.

On the held-out test set, the classifier achieved an accuracy of 0.920, an efficiency of 0.975, a purity of 0.879, and an AUC of 0.985. The confusion matrix yielded 1,462 true positives, 1,299 true negatives, 38 false negatives and 201 false positives, indicating that the classifier is somewhat more prone to accepting spurious noise events than to missing genuine neutrino events.

The effect of varying the classification threshold is shown in Figure 7. At the default threshold of 0.5, the operating point sits in a region of balanced efficiency and purity. Lowering the threshold to 0.3 increases efficiency to 0.977 at the cost of reducing purity to 0.843, reflecting a shift towards higher neutrino detection completeness. The ROC curve confirms strong discriminative power across all thresholds, with the AUC of 0.985 indicating near-ideal separation between the two classes at $\sigma = 5$.

| | **Predicted Signal** | **Predicted Noise** |
|---|---|---|
| **True Signal** | 1462 (TP) | 38 (FN) |
| **True Noise** | 201 (FP) | 1299 (TN) |

Table 3: Confusion matrix at $\sigma = 5$.

| Metric | Threshold = 0.5 | Threshold = 0.3 |
|---|---|---|
| Accuracy | 0.9203 | 0.8977 |
| Efficiency | 0.9747 | 0.9773 |
| Purity | 0.8791 | 0.8430 |
| FPR | 0.1340 | 0.1820 |
| AUC | 0.9850 | – |

Table 4: Classification performance at $\sigma = 5$ under two classification thresholds. AUC is threshold-independent and is reported once.

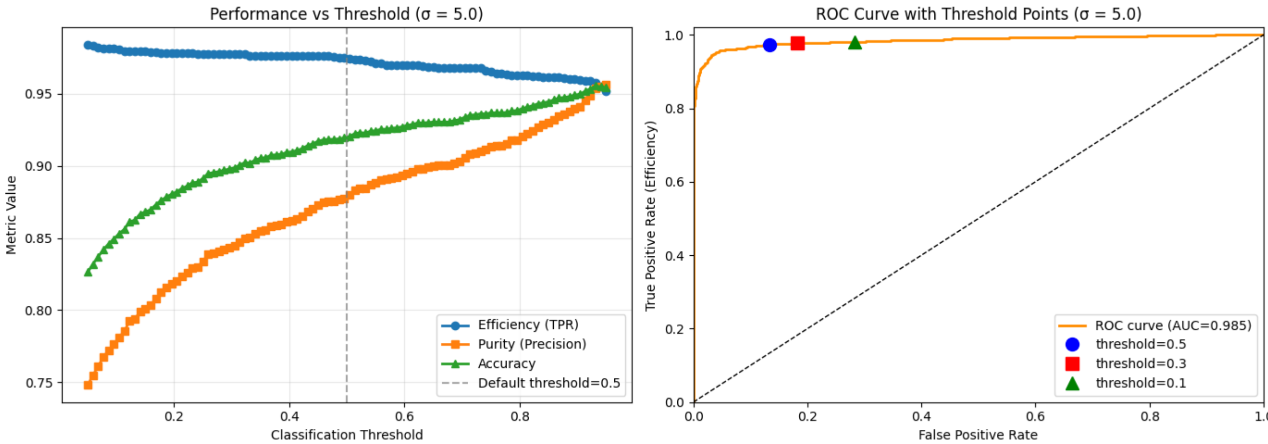
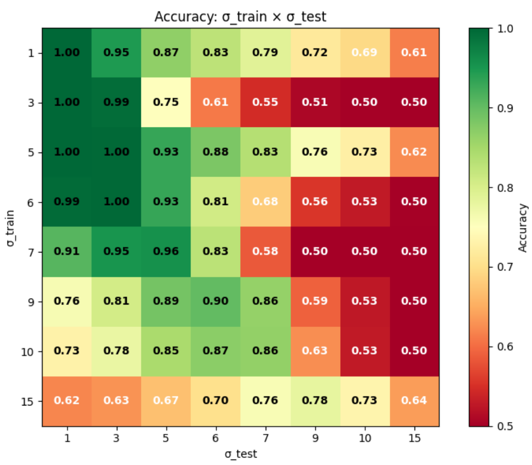


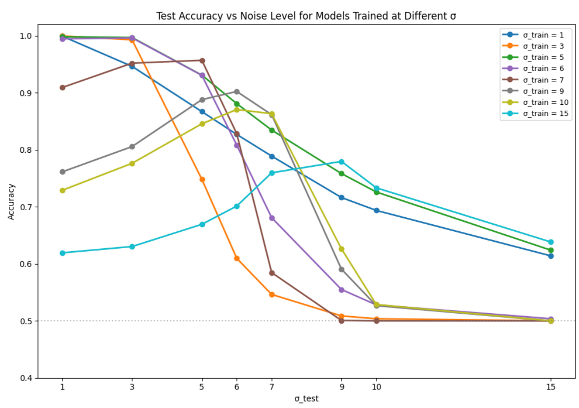

Figure 7: Threshold analysis for the baseline classifier at $\sigma = 5$. Left: efficiency, purity, and accuracy as a function of classification threshold. Right: ROC curve with operating points at thresholds 0.1, 0.3, and 0.5 indicated.

### 3.2 Electronic Noise Sweep

The $8 \times 8$ accuracy matrix across all combinations of $\sigma_{\text{train}}$ and $\sigma_{\text{test}}$ is shown in Figure 8a. The model trained at $\sigma_{\text{train}} = 5$ achieved the highest mean accuracy of 0.844 across all test conditions. Performance generally degraded as $\sigma_{\text{test}}$ increased beyond the training noise level, with most models approaching random classification at $\sigma_{\text{test}} = 15$.




(a) 8×8 accuracy matrix.




(b) Test Accuracy vs Noise Level curves.

Figure 8: Electronic noise sweep results showing (a) accuracy heatmap for all training and testing noise combinations and (b) corresponding accuracy curves highlighting trends across noise levels.

The accuracy curves in Figure 8b revealed distinct generalisation behaviours across training conditions. The model trained at $\sigma_{\text{train}} = 3$ collapsed rapidly beyond its training condition, falling to an accuracy of 0.75 at $\sigma_{\text{test}} = 5$ and near-random performance at $\sigma_{\text{test}} \geq 9$, indicating that low-noise training produced features insufficiently robust to higher noise levels. A similar but sharper collapse was observed for $\sigma_{\text{train}} = 7$, which achieved a peak accuracy of 0.96 at $\sigma_{\text{test}} = 5$ before dropping abruptly to 0.58 at $\sigma_{\text{test}} = 7$. Models trained at high noise levels, such as $\sigma_{\text{train}} = 9$ and $\sigma_{\text{train}} = 10$, showed the opposite trend, performing poorly at low $\sigma_{\text{test}}$ but maintaining moderate accuracy across the mid-range.

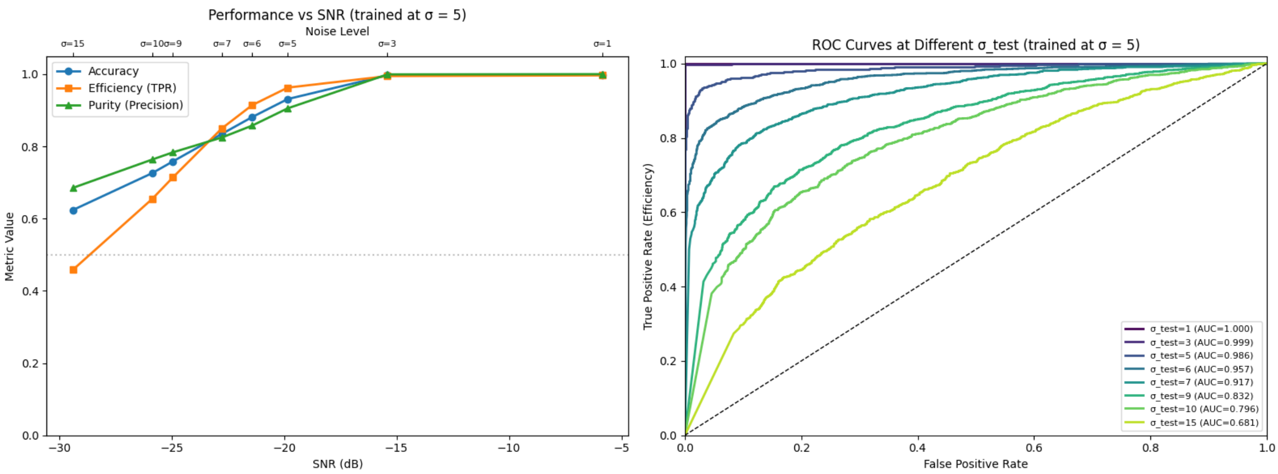



Figure 9: Performance of the $\sigma_{\text{train}} = 5$ model as a function of signal-to-noise ratio (SNR), and $\text{SNR} = 10\log_{10}\left(\frac{x^2}{\sigma^2}\right)$ dB. Left: accuracy, efficiency, and purity versus SNR. Right: ROC curves at each $\sigma_{\text{test}}$ value, with corresponding AUC scores indicated in the legend.

The performance of the $\sigma_{\text{train}} = 5$ model as a function of the signal-to-noise ratio (SNR), defined as the ratio of mean signal power to mean noise power expressed in decibels, is shown in Figure 9. All three metrics remained above 0.83 up to $\sigma_{\text{test}} = 7$ (SNR $\approx -23$ dB), beyond which accuracy and efficiency declined notably, identifying this as the practical degradation threshold. The ROC curves confirmed progressively weaker discriminative power at higher noise levels, with AUC falling from 0.999 at $\sigma_{\text{test}} = 3$ to 0.681 at $\sigma_{\text{test}} = 15$.

### 3.3 Radioactive Noise Classification

The classifier maintained near-perfect performance across a wide range of radioactive noise intensities. Accuracy, efficiency and purity all remained above 0.995 for $\lambda \leq 100$, and performance remained strong at $\lambda = 500$ with an accuracy of 0.958. A sharp collapse occurred between $\lambda = 500$ and $\lambda = 1000$, where accuracy fell to 0.499 and efficiency to 0.457, corresponding to near-random classification.

The direct comparison between radioactive and electronic noise performance is shown in Figure 10. At all tested values of $\lambda$ up to 500, the radioactive noise classifier substantially outperformed the electronic noise baseline at $\sigma = 5$, demonstrating that the classifier is considerably more robust to radioactive noise than to electronic noise at comparable difficulty levels. The classifier tolerates up to two orders of magnitude more blobs per image before performance degrades meaningfully. The failure mode at $\lambda = 1000$ is characterised by a simultaneous drop in both efficiency and purity, indicating that the classifier loses discriminative ability entirely rather than developing a systematic bias towards one class, a pattern qualitatively distinct from the gradual efficiency-led degradation observed for electronic noise.

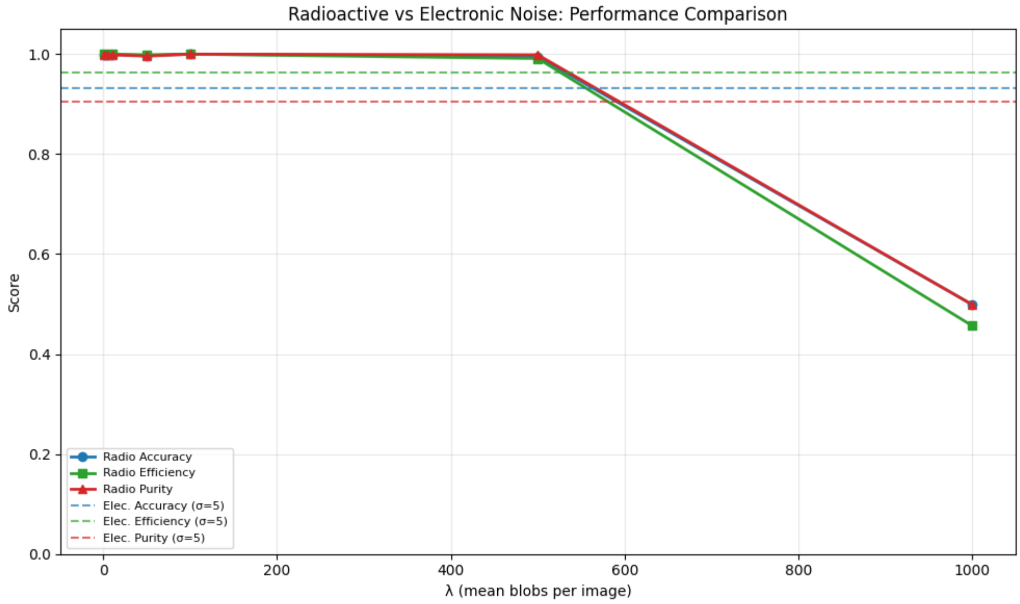



Figure 10: Direct comparison of classifier performance under radioactive (solid lines) and electronic noise (dashed lines, $\sigma = 5$) as a function of $\lambda$. Accuracy, efficiency, and purity are shown for each noise type.

## 4 Discussion

The superior generalisation of the model trained at $\sigma_{\text{train}} = 5$ is physically motivated. At this noise level, the electronic noise standard deviation is comparable to the mean non-zero pixel value of the neutrino images, forcing the network to learn spatially structured track features. Models trained at lower $\sigma$ instead learned features tied to near-clean images, which proved insufficiently discriminative as noise increased, producing the sharp generalisation collapses evident in Figure 8b. The contrasting failure modes reflect different decision boundaries induced by the respective training distributions. The $\sigma_{\text{train}} = 3$ model preserved purity while efficiency collapsed, indicating excessive conservatism, whereas the $\sigma_{\text{train}} = 1$ model maintained relatively higher efficiency at the cost of purity, suggesting a more permissive decision boundary that admitted a greater fraction of noise events as signal.

| $\sigma_{\text{test}}$ | Accuracy | Efficiency | Purity |
|---|---|---|---|
| 1 | 0.999 | 0.999 | 1.000 |
| 3 | 0.946 | 0.992 | 0.909 |
| 5 | 0.867 | 0.939 | 0.821 |
| 6 | 0.827 | 0.891 | 0.790 |
| 7 | 0.789 | 0.835 | 0.764 |
| 9 | 0.716 | 0.737 | 0.708 |
| 10 | 0.694 | 0.703 | 0.690 |
| 15 | 0.614 | 0.582 | 0.622 |

$\sigma_{\text{train}} = 1$

| $\sigma_{\text{test}}$ | Accuracy | Efficiency | Purity |
|---|---|---|---|
| 1 | 1.000 | 0.999 | 1.000 |
| 3 | 0.993 | 0.989 | 0.997 |
| 5 | 0.748 | 0.497 | 1.000 |
| 6 | 0.610 | 0.219 | 1.000 |
| 7 | 0.546 | 0.093 | 1.000 |
| 9 | 0.509 | 0.017 | 1.000 |
| 10 | 0.504 | 0.007 | 1.000 |
| 15 | 0.500 | 0.000 | 0.000 |

$\sigma_{\text{train}} = 3$

Table 5: Classification performance across test noise levels for models trained at $\sigma_{\text{train}} = 1$ (left) and $\sigma_{\text{train}} = 3$ (right), illustrating the contrasting failure modes of low-noise training.

The greater robustness to radioactive noise is attributable to the spatial sparsity of blob-based backgrounds. Even at large $\lambda$, individual blobs occupied a small fraction of the image, leaving the localised track structure of neutrino events partially visible. Electronic noise, by contrast, simultaneously introduced fluctuations across the entire image, globally obscuring the signal and explaining the earlier onset of performance degradation. The abrupt collapse at $\lambda = 1000$ reflected a transition in which background images became statistically indistinguishable from signal images at very high blob densities.

From the perspective of supernova neutrino detection, missed signal events represent irreversible observational losses, whereas false positives are recoverable through subsequent analysis. This asymmetry motivates operating below the default threshold of 0.5 to prioritise efficiency, as demonstrated by the threshold scan in Section 3.1.

# 5 Conclusion

A CNN was trained to classify simulated LArTPC images as neutrino-plus-noise or pure noise, achieving an accuracy of 0.920 and an AUC of 0.985 at the physical baseline of $\sigma = 5$. A systematic sweep across noise levels identified $\sigma_{\text{train}} = 5$ as the optimal training condition, with reliable performance maintained up to $\sigma_{test} \approx 7$. The alignment between the physically motivated baseline and the empirically optimal training level indicates that noise conditions comparable in scale to the signal are most conducive to learning robust features. Under radioactive noise, near-perfect classification was maintained up to $\lambda = 500$, with the classifier tolerating two orders of magnitude more background than in the electronic noise case, a robustness attributed to the spatial sparsity of blob-based deposits.

These results demonstrate the feasibility of CNN-based supernova neutrino detection in liquid argon detectors and highlight the importance of noise shape in determining classifier robustness. Future work should consider more realistic noise models and validation on actual LArTPC detector data.

# References

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